CIVE 202 – Civil Engineering Analysis II

Project #4 Report

Spring 2026

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Introduction/Background Information

The Federal Emergency Management Agency (FEMA) developed the National Risk Index (NRI) as a comprehensive tool to measure and compare natural hazard risk across different communities in the United States. Risk is defined as the product of probability of an event times impact of the event. The National Risk Index provides risk percentiles and ratings based on data for Expected Annual Loss due to natural hazards, Social Vulnerability, and Community Resilience. These components produce a Risk Index that ranks geographic areas based on percentile scores and quality ratings that range from "Very low" to "Very high." The NRI is widely used in the United States to identify areas most at risk of a natural disaster. However, like many models, they often have potential biases that can influence how risk is represented. These biases arise from different factors including the weighting of variables, selection of datasets or the way that social/economic factors are measured. As a result, different definitions of risk may yield different conclusions about the communities with most vulnerability.

Risk Averse, LLC, an independent risk analysis consulting research group, has asked our team to perform a sensitivity analysis that can review where biases may exist within the NRI scoring method. Additionally, they have asked us to create an alternative risk scoring method that can use the NRI dataset as an input. After creating this, we will compare the NRI to our risk index and determine if there are any categorical biases in how risk in scored for both methods.

Objectives

The following are the objectives of this analysis:

- To clean and organize the Census Tract level dataset from NRI, as well as the Social Vulnerability Index (SVI) dataset

- To merge multiple datasets together from the NRI database
- To identify potential biases within the NRI scoring method and analyze the dataset provided
- To create a method different from the NRI method for analysis natural disaster risk in different areas
- To generate two summary tables or figure showing risk scoring by the NRI method and our own method
- To generate a map using GeoPandas that communicates risk information

Scope of Work

Team Members: Gabriela Gonzalez Ng, Cam Becker, Gavin Rocha, Ashlee Rodriguez

Date submitted for review: April 21st, 2026

Client: Risk Averse, LLC

Submitted to: Civil and Environmental Engineering Department faculty

Project Goals:

The main goal for this project is to evaluate and analyze natural disaster risk in two states using the National Risk Index (NRI) data from the Federal Emergency Management Agency's (FEMA) and the Social Vulnerability Index (SVI) data from ASTDR. The NRI risk score is calculated by combining the Expected Annual Loss with Social Vulnerability and based on this risk score a higher score indicates a greater relative risk. The two states we will be examining are Ohio and Missouri. We will assess whether potential categorical biases exist within its current risk scoring method. The client has requested three key analyses: (1) compare the current risk levels determined using the NRI method and data to an independent consultant's risk. (2) create an alternative risk scoring method that can use the NRI dataset as an input. (3) compare NRI to our risk index to determine categorical biases.

Project Tasks:

To achieve the goals for our project, our group has identified the following tasks to be completed by our team:

Task 1: Download, organize, clean and merge NRI and SVI datasets

Data for the NRI and SVI dataset will be extracted and organized in Jupyter notebook to process in the Python software. This data will be joined with the SVI dataset to aid Risk Averse, LL. Missing or NA data will be identified and handled by removing from the dataset to ensure accuracy in the analysis.

Task 2: Create an alternative risk scoring method

An alternative risk scoring method will be developed to provide an independent evaluation of natural disaster hazards. This will include research on how risk is calculated and analysis of key components that affect risks. We will compare this alternative method to the NRI and determine any categorical biases within both risk methods.

Task 3: Generate summary tables or figures

For visualization purposes, summary tables and figures will be created for both the NRI and our proposed risk scoring method. These visualizations will highlight key trends and differences in risk levels across the selected states.

Task 4: Produce GeoPandas map

A map will be created to communicate risk information to the client. This will be done using the GeoPandas library in Python.

Project Deliverables:

The deliverables and due dates for this project are summarized in the following table:

Deliverable	**Due to Client by**
Raw Data files	4-21-26
Python Code File	4-21-26
Annotated Code Document	4-21-26
Written Report	4-21-26
Scope of Work	4-21-26
Engineering Timesheet	4-21-26
Gantt Chart	4-21-26
GitHub Repository link	4-21-26

Methods

In order to identify biases in the NRI dataset and create our own method of risk scoring, our team used the following methods in chronological order:

Data Source

The Census Tract level for Ohio and Missouri was provided by the Department of Homeland Security. This dataset included information about the risk of the natural hazard in the population in an area such as Expected Annual Loss and Hazard Exposure. The Social Vulnerably Index (SVI) ranks communities based on their vulnerability to disasters and includes data such on things such as socioeconomic status. All the analyses conducted were made using these two CSV files.

Data Extraction and Organization

Data was extracted in Jupyter notebook (Python version 3.14.2) by importing the pandas' package and converting it into a dataframe. Additionally, the Seaborn library was installed to create the visualizations of the CSV data provided. Data was organized into tables and filtered for easier analysis. GeoPandas was installed and used to create a map showing risk. The two datasets were merged in order to facilitate analysis.

Data Cleaning

To ensure the right accuracy, the following things were done:

- **Handing missing values**: Cells with missing values were removed or imputed using mean/median substitution.
- **Data type corrections:** Variables were properly formatted.
- **Outlier assessment:** Values that were extreme were identified and reviewed to determine if they reflected real – world conditions.

Sensitivity Analysis

A sensitivity analysis was conducted to evaluate how changes in the NRI model and our model inputs affect risk outputs. This included replacing missing values with mean or median values and adjusting the weights assigned to the variables used in the NRI risk formula such as Expected Annual Loss, Social Vulnerability, and Community Resilience. The results of this demonstrated biases that exist inside of the NRI or alternative model.

Data Visualization

Summary tables and visualizations were generated to compare the difference in risk code between the two methods. Geospatial map was created using GeoPandas to illustrate the spatial distribution of risk across Ohio and Missouri. Two maps were created based on the NRI risk scores and the alternative model, to allow for identification of potential patterns in risk classification and definition.

Comparison

To identify potential biases in the risk score methods, the NRI and the alternative model were compared at the Census Tract level. Differences in categories were analyzed to identify certain patterns. Additionally, communities where compared with those of high social vulnerability or low resilience, to indicate whether those communities were consistently assigned higher or lower risk scores depending on the model used. These differences were used to identify the presence of biases in the NRI model.

Results and Discussion

Based on the analysis conducted for the NRI and SVI data, our team of engineers have found the following conclusions:

Tornados Vs. Landslides

When comparing tornados vs. landslides in Ohio and Missouri, there were multiple patterns found. Missouri's comparison shown in *Figure 1* showed that the risk for tornados in different counties was relatively high, and common for multiple counties. However, when looking at the landslides risk, there were common outliers in the data found as some counties had a high mean of landslide risk, while others had low mean risk or none at all. Similarly, Ohio in *Figure 2* showed varied tornado risks with a group of counties showing high tornado risks while others low. For the landslide risk, there were common outliers, however it was a more evenly distributed across the state in comparison to Missouri's landslide risk.

This pattern indicates that different hazards can exhibit distinct risks in different regions. Additionally, it is important to note that mean values can be useful for comparing the overall risk levels between states, however it is important to note that it might not truly capture a state's risk level for natural hazards. The dataset might contain outliers which affect the average risk levels as it can inflate the mean. This demonstrates that biases might exist within a model depending on how data is summarized, and it's important to incorporate additional measures to accurately analyses and interpret the data.

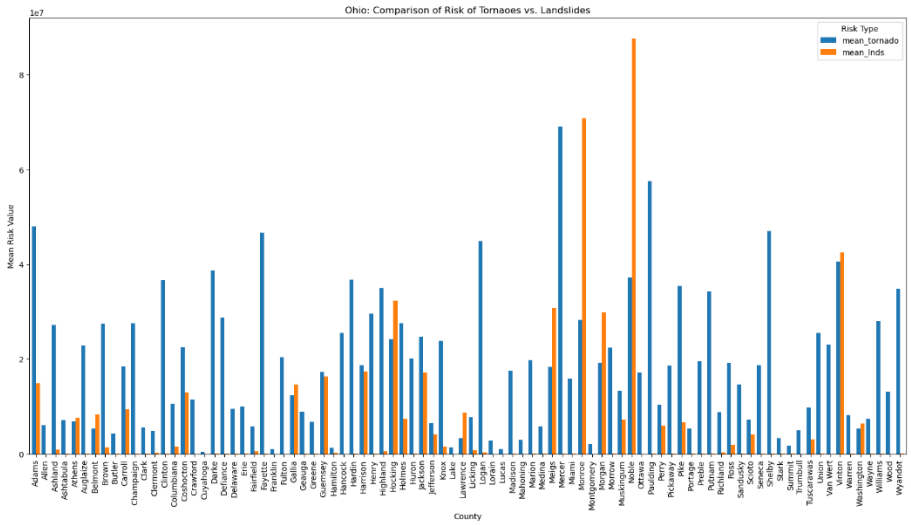

Figure 1 Ohio Risk of Tornadoes vs. Landslides

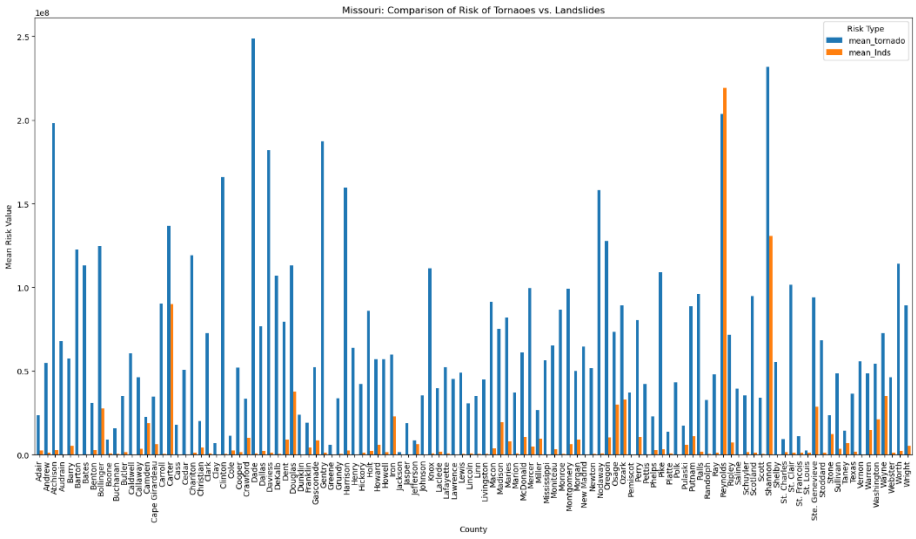

Figure 2 Missouri Risk of Tornadoes vs. Landslides

GeoPandas

To visualize the differences between risk in Ohio vs. Missouri, GeoPandas were used to create geospatial maps.

The results showed that for tornado risk in *Figure 3 and Figure 4*, Missouri had a higher chance of tornadoes as it showed most areas had a risk score of 70-100. Whereas for Ohio, the areas on the west side showed a high-risk score for tornadoes, but the areas closer to the southeast showed risk levels for tornadoes low, with some showing risk levels of under 20. This correlates with known regional hazard patterns, as Missouri lies within the Midwest region, an area that is frequently impacted by tornados. In contrast, Ohio has elevated levels of risk for tornadoes on the west side since it is closer to the Midwest region, however Ohio experiences less consistent and lower-intensity events compared to Missouri.

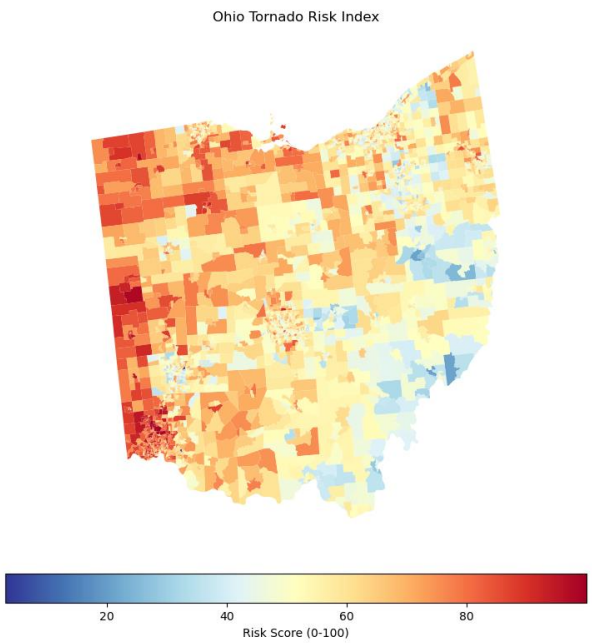

Figure 3 Ohio Tornado Risk Index

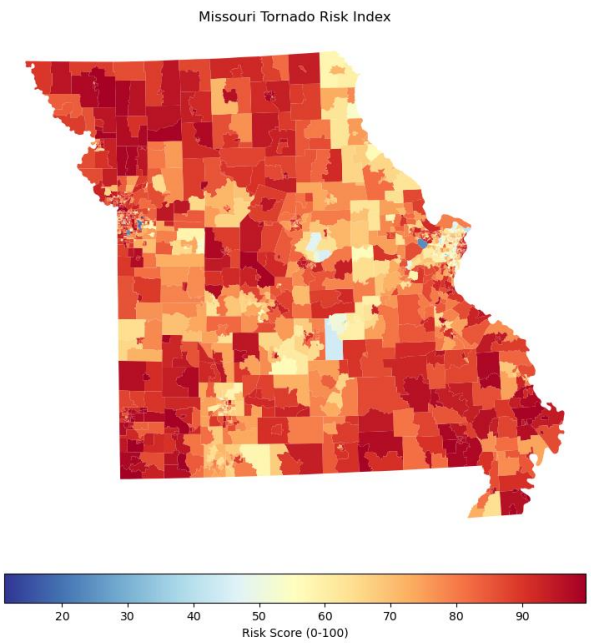

Figure 4 Missouri Tornado Risk Index

When comparing Landslide risk, Missouri had a consistent risk for landslides, showing risk levels ranging from 60-100 with some outliers near the southside of the state. In comparison, Ohio showed high risk levels of landlide in the southeast region and varied risk levels to the northwest. This pattern reflects the geographical differences between the two

states. Missouri consistent landslide risk suggest that there are factors such as soil compostion and slope conditions affecting the risk for landslide. Ohio's highest landslide risk occurring in the southeaster portion of the state align with the presence of the Appalachian Plateu, which include more erosion-prone soils and steeper slopes.

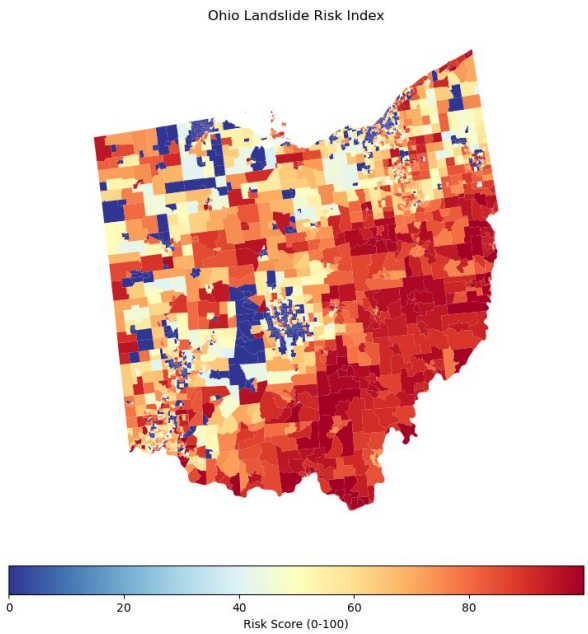

Figure 5 Missouri Landslide Risk Index

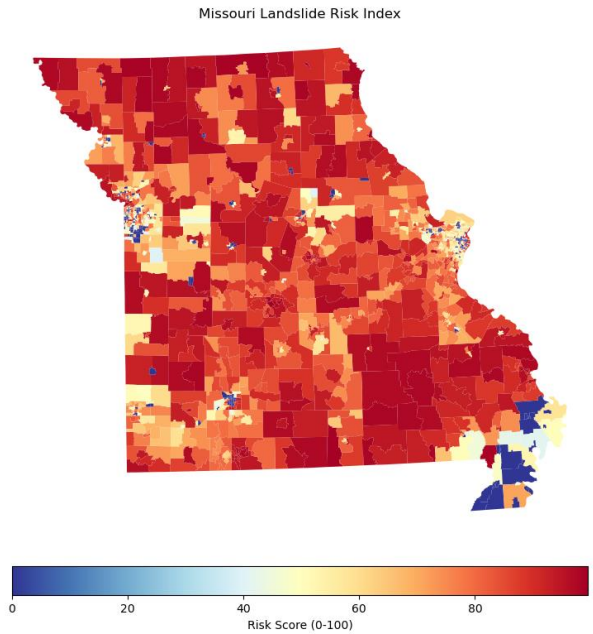

Figure 6 Ohio Landslide Risk Index

Alternative Risk Analysis

The alternative risk model that our team developed calculated risk using a different approach from the National Risk Index (NRI). The model that we developed uses the following formula:

$$Risk = \frac{EAL * Social\ Vulnerability}{Community\ Resilience}$$

By using this formula creates a relationship that amplifies risk in areas where hazard exposure and social vulnerability are high. Dividing by resilience would add an inverse effect, as areas with lower resilience will experience higher risk values, while areas with higher reselience reduces the risk of the natural hazard.

Comparison between models

The comparison between the National Risk Index (NRI) and the alternative risk model relvealed noticeable differences in how risk is scored across the Census Tracts in Ohio and Missouri. In contrast to the alternative model, the NRI uses a composite scoring system that converts EAL, SVI, and resillience into percentil rankings and combines them to create a risk index. When comparing them the alternative model tended to produce higher risk values in areas with high social vulnerability and lower resilience. On the other hand the NRI produced moderate risk classifications that were more evenly distributed across regions. This means that the alternative model is sensitive to extreme values and specific regions. The results showed that although both models incorporate similar variables, the used of raw values vs percentile aggregation can influece how risk is represented and difined. This made lead to different interprétations of community vulnerability and risk of natural disasters.

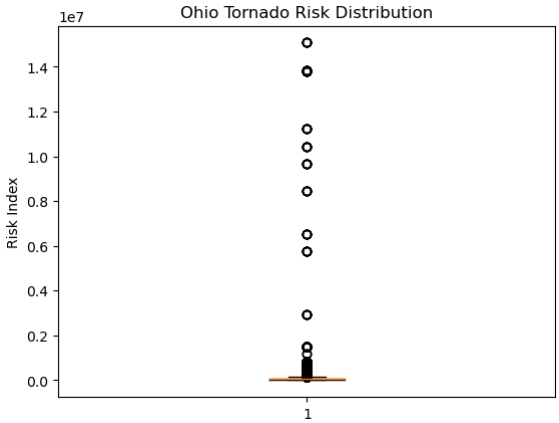

Figure 7 Alternative Model Ohio Tornado Risk

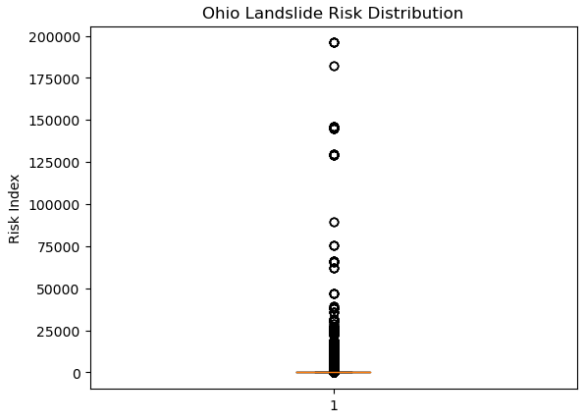

Figure 8 Alternative Model Ohio Landslide Risk

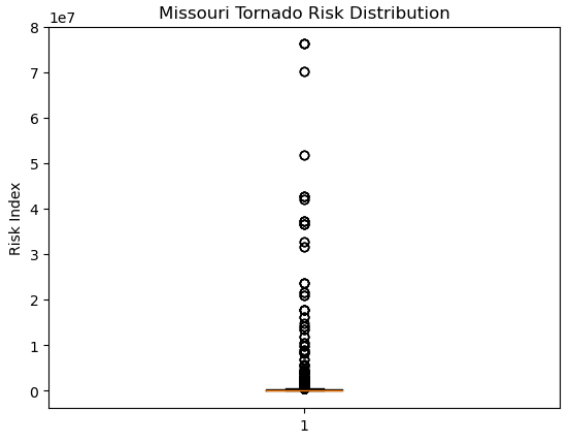

Figure 9 Alternative Model Missouri Tornado Risk

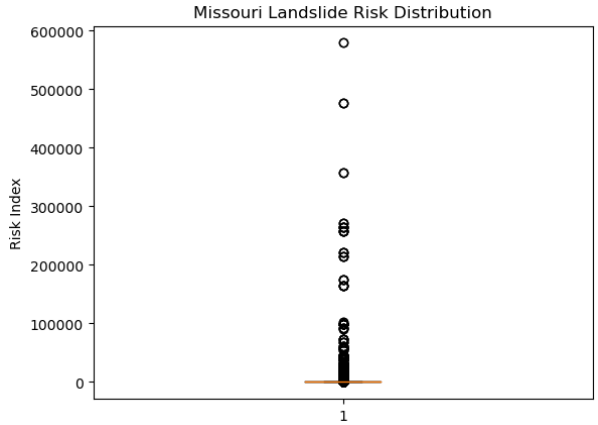

Figure 10 Alternative Model Missouri Landslide Risk

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